# ResGate: Topic-Scoped Refusal Removal via Single-Layer Residual Gating

Kai Ni

Draft, 15 October 2026

**Abstract**

A transformer computes layer by layer through a residual stream, and that ordering can be put to work. We confine a rank-1 refusal-removing weight edit to a contiguous **band** of middle layers — 6 tensors at 9.4% of depth on Phi-4-mini — and read a binary gate from a layer **strictly below** that band. Every layer the gate reads is therefore left untouched, so the hidden state at the read site is **bit-identical whether or not the edit will fire**. The gate's decision is causally prior to the edit, and **one forward pass serves a conditional edit**: no second pass, no speculative generation, no router trained against the edited model. Band surgery is what creates the room. An edit spread over all depth would leave nowhere to read from.

The gate is closed form throughout — one direction and one threshold, computed from activations at one layer by subcategory-centred within-class scatter and smooth Tikhonov shrinkage, with no gradients, no search and no learned router. On three architectures the edit removes refusal almost completely at low KL (0.0745–0.0905), and the gate withholds it: on every prompt where the gate does not fire the gated output is **byte-identical** to the unedited model, while the same edit applied unconditionally changes every one of those outputs. The gate routes **78%, 81% and 82%** of 675 fresh out-of-corpus cyber prompts to the edit (lifting judged compliance on a 64-prompt fresh cyber sample to **85.9% on all three models**) and fires on **7%, 9% and 7%** of 234 non-target harmful prompts. On the prompts the unedited model refused, the gate bit and the judged outcome agree on **294 of 294**: nothing was answered without the gate firing, and nothing that fired was refused. The same edit with no gate answers 100% of those prompts, so the gate withholds a capability that is fully present. A second target category, bio, reaches **98.9–99.4%** coverage (**127/127 = 100%** on held-out validation rows) at **0.0–0.4%** leak onto the cyber set and **0/637** onto same-corpus cyber prompts.

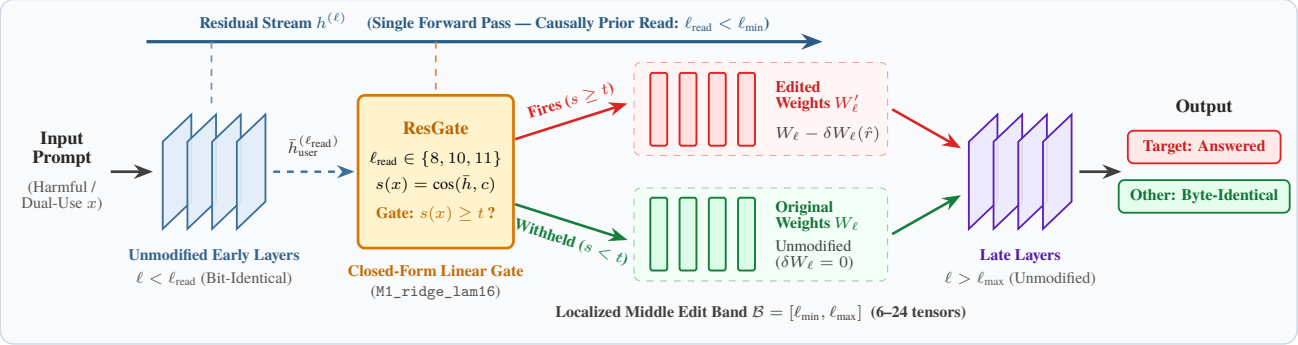



**Figure 1: ResGate single-pass architecture.** An input prompt flows through unmodified early layers ($\ell < \ell_{\text{read}}$) along the residual stream. At $\ell_{\text{read}}$, **ResGate** pools user-turn activations $\bar{h}_{\text{user}}^{(\ell_{\text{read}})}$ and evaluates a closed-form linear gate $s(x) = \cos(\bar{h}, c) \geq t$ (`M1_ridge_lam16`). Because $\ell_{\text{read}}$ is strictly below the localized middle edit band $\mathcal{B} = [\ell_{\min}, \ell_{\max}]$, the hidden state at the gate is causally prior to the edit. If the gate **fires** (target topic), rank-1 refusal-removing weight edits $W'_\ell$ are applied across $\mathcal{B}$; if **withheld** (non-target harm or benign prompt), computation proceeds through the **original weights** $W_\ell$, producing an output byte-for-byte identical to the unedited model in a single forward pass.

## 1 Introduction

Abliteration removes a language model's refusal behaviour with a rank-1 orthogonalising weight edit [4]. Doing so for one harm category while preserving refusal elsewhere has been approached by shaping the contrast dataset, which leaves the edit permanent and always-on. We make the edit conditional instead, and the mechanism is a **separation of layers**.

A decoder-only transformer writes into a residual stream layer by layer. Confine the edit to a contiguous band of middle layers and every layer *below* the band is left untouched — not approximately, but bit for bit, in both arms. A linear classifier read from one of those layers therefore sees the same hidden state whether or not the edit will fire downstream. Its decision is causally prior to the edit, so it can be taken once

at prefill and the edit applied, or withheld, in the *same* forward pass. Three things are doing work and they are not interchangeable: the architecture is what *permits* this, band surgery is what *creates the room*, and the gate is what *uses* it.

That ordering also makes the design verifiable in a way a conditional intervention usually is not. If the gate's input cannot depend on the edit, then on a prompt the gate declines the served output must be the unedited model's output exactly — and it is, byte for byte, on every withheld prompt of two harmful test sets and three architectures (§6.3). The literature's verdict on gates is that the gate is the fragile part [11, 5, 33]; byte identity is the strongest answer we can give to it.

### 1.1 Contributions

- **Route-invariant read-early, edit-late gating.** Reading the gate strictly below the edit band ensures that $h$ at the read site is identical whether or not the edit fires, so a single forward pass serves a conditional weight edit with exact byte identity on withheld prompts (§2.4; building on the early-read routing pattern of Residual Paving [11]).
- **A closed-form ridge gate.** Residual Paving [11] fits its router by gradient descent with weighted BCE and identifies learned route selectivity as its primary bottleneck (an oracle router gains a median $+12.9$ pp across six backbones, up to $+50.7$ pp). Our gate direction, edit band, and threshold are instead computed in closed form from activations via subcategory-centred scatter and Tikhonov shrinkage (§2.3): no gradients, no search, no learned router, and no random seed.
- **Gated rank-1 orthogonalisation.** Coupling a pre-band linear gate with a rank-1 row-normalised weight edit yields targeted compliance on the selected category while leaving non-target refusals intact, combining three properties that have previously appeared only in pairs (§3).
- **Live end-to-end verification.** Across three architectures and two target domains (cyber and bio), we verify exact byte identity on withheld prompts and a 294-of-294 agreement between the gate bit and the judged outcome on refused prompts.

**Status of the figures in this paper.** Every figure below is measured and traceable to a record directory (§9). Throughout, **LIVE** means served by our own HTTP server with one real forward pass per prompt and **OFFLINE** means scored from stored activations; where a quantity was not measured we say so and print no number for it.

## 2 Method

### 2.1 The rank-1 residual-stream edit

For each matrix $W$ that **writes to the residual stream** — `self_attn.o_proj`, `mlp.down_proj`, and `experts.down_proj` on MoE models — and a unit refusal direction $v$ in output space (orthogonalised against the harmless centroid following [4, 14]), the rank-1 PRE row-normalised update is $W' = W - w\,(n \odot v)\,u^\top$ with $n = \text{rownorm}(W)$ and $u = v^\top(W/n)$. Row-normalising before forming the pullback keeps the correction uniform across output rows of very different scale (reproducing the unconstrained reference implementation [14] to a cosine of $0.99985$).

Rather than materialising modified weights on disk, we serve the equivalent forward hook

$$y' = y - w\,(x \cdot u)\,(n \odot v), \tag{1}$$

which matters for §6.3: on a prompt the gate withholds, this subtracts nothing, so the computation is the unedited model's own and the base weights are never touched. Query, key and value projections are left alone — they read the residual stream, they do not write it.

The single-direction premise the edit rests on is Arditi et al. [4]; for published arguments that refusal is not exhausted by one direction see Joad et al. [15], Wollschläger et al. [31] and Piras et al. [23].

### 2.2 Localising the edit band

To place the gate strictly below the intervention, we first localise the edit to a contiguous mid-network band. The direction at layer $l$ is the difference of means of harmful and harmless activations at the last prompt token, orthogonalised against the harmless centroid; on why a pure through-origin direction can be the wrong object see Marshall et al. [19]. The **dose weight** is an effect size, not a raw gap:

$$\text{sep}(l) \;=\; \frac{\left\|\text{mean}_i(h_{i,l}) - \text{mean}_j(g_{j,l})\right\|}{\text{mean}_j\left\|g_{j,l} - \bar{g}_l\right\|}, \tag{2}$$

the harmful-to-harmless gap measured **in units of the harmless cloud's own radius**. Dividing by the spread removes the part of the gap that merely reflects a layer having large activations. The weights are normalised to mean 1, so the scalar $K$ sets total edit energy and $\text{sep}(l)$ only redistributes it: $w_l = K \cdot \text{sep}(l)/\,\text{mean}_l\,\text{sep}(l)$. The quantity to dose-match across models is not $K$ but $\max|\Delta W|$, which is linear in $K$.

The band is **measured per model**, never assumed: compute $w(l)$ for every layer from one capture, pooling at the last prompt token; admit the layers with $w(l) \geq 0.9 \cdot \max_l w(l)$; take the contiguous hull of that set. The bands so found span 7 of 30, 3 of 32 and 12 of 32 layers — **23.3%, 9.4% and 37.5% of depth**. The ratio of largest to smallest $w$ across depth is $\approx 3.6$, 2.43 and 2.17.

**The rule finds a peak on two models and a plateau on the third.** Phi's $w$ is 1.481, 1.482, 1.393 against a

model maximum of 1.482 — unambiguous, with the peak at L14. Llama's sits between 1.10 and 1.29 from L11 all the way to L27, so the admitted set is non-contiguous ($\{11, 12, 13, 15, 16, 20, 21, 22\}$) and the hull adds four layers purely to close it. We report Llama's band as a weakness of the model, not of the rule, and "surgical" is a far weaker claim at 37.5% of depth than at Phi's 9.4%. Contiguity is a specification convenience, not a requirement: the implementation subsets the band at run time, so any set of layers works.

**What the band buys is not only KL.** A narrow band is usually motivated as a cheaper edit. Its load-bearing property here is different: it leaves a majority of the depth — and in particular the layers below it — *unmodified*, which is the only reason a read can be placed where the edit cannot reach it. Where refusal lives in depth is contested, and we cite that rather than settle it: Zong et al. [37] find the block spanning layers 8–11 selected first in all six of their greedy searches; Eswar et al. [7] find a *"redundant subspace"* admitting many disjoint sufficient masks; Li et al. [18] read a contiguous mid-band the other way, as *necessary* "safety layers"; and Hase et al. [9] show that causal-localisation conclusions *"do not provide any insight into which model MLP layer would be best to edit"*.

## 2.3 The gate: how it is extracted

The gate is **binary**: one direction, one threshold, one bit, frozen at prefill. For a target category against all other harm categories the between-class scatter has rank $C - 1 = 1$, so one direction is the correct object rather than a simplification.

**Notation.** All rows are pooled over the user-content span only (`mean_user`) at read layer $\ell$, and **each row is unit-normalised before any mean**: $X_p$ are the TRAIN rows of the target category; $X_n$ the REF rows, 10 drawn from each non-target harm category; $p = \text{mean}(X_p)$ and $g = \text{mean}(X_n)$; $d = p - g$; and $n = \text{rows}(X_p) + \text{rows}(X_n)$.

**Subcategory-centred within-class scatter.** The raw difference of means $d = p - g$ points partly along directions that are wide *within* each class — nuisance variance such as prompt length, syntactic framing and register. Because the reference set $X_n = \bigsqcup_{s=1}^{S} X_{n,s}$ spans $S$ distinct non-target harm categories ($S = 14$ categories of $n_s = 10$ rows each on the 240-row fit), centring $X_n$ on a single pooled mean $g = \text{mean}(X_n)$ would leave the $S - 1$ between-subcategory offsets $(\bar{g}_s - g)$ inside the scatter matrix, where $\bar{g}_s = \text{mean}(X_{n,s})$:

$$\Sigma_{\text{global}} = \Sigma_{\text{subcat}} + \frac{1}{n-2} \sum_{s=1}^{S} n_s (\bar{g}_s - g)(\bar{g}_s - g)^\top . \quad (3)$$

Those $S - 1$ inter-category offsets inflate the leading covariance modes and crowd out true within-category nuisance variance. We therefore centre the target class on $p$ and each negative subcategory $X_{n,s}$ on its own centroid $\bar{g}_s$, removing between-subcategory scatter by construction while keeping the mean-difference signal $d = p - g$ unchanged:

$$\begin{aligned} Z_{\text{subcat}} &= \big[\, X_p - p \;;\; X_{n,1} - \bar{g}_1 \;;\; \ldots \;;\; X_{n,S} - \bar{g}_S \,\big], \\ \Sigma_{\text{subcat}} &= \frac{Z_{\text{subcat}}^\top Z_{\text{subcat}}}{n - 1 - S}. \end{aligned} \quad (4)$$

Note that $\Sigma_{\text{subcat}}$ is **within-class**, not total, covariance — this is the LDA object, not PCA. Centring each class and subcategory on its own mean first removes the between-class signal, so the whitening suppresses nuisance rather than the thing we want to read.

**Smooth Tikhonov whitening via the Gram spectrum.** Rather than truncating the covariance spectrum at a hard top-$k$ step (which leaves 29–50% of $\|d\|^2$ across modes $6 \ldots 15$ unwhitened), we apply smooth Tikhonov shrinkage anchored at an eigenvalue scale $\alpha = \lambda_{k+1}$ of $\Sigma_{\text{subcat}}$:

$$\begin{aligned} c &= \text{unit}\big(\alpha(\Sigma_{\text{subcat}} + \alpha I)^{-1} d\big) \\ &= \text{unit}\Big( (I - VV^\top)\, d + V\, \text{diag}\Big(\frac{\alpha}{\lambda_i + \alpha}\Big) V^\top d \Big), \end{aligned} \quad (5)$$

where $(\lambda_i, v_i)$ are all positive eigenpairs of $\Sigma_{\text{subcat}}$. Because $\text{rank}(\Sigma_{\text{subcat}}) \le n - 1 - S$, the $d \times d$ matrix is never formed; the eigenvectors $V$ are lifted in closed form from the $n \times n$ Gram matrix $Z_{\text{subcat}} Z_{\text{subcat}}^\top$, whose symmetry and positive semi-definiteness make the lifted columns orthonormal — load-bearing, since $VV^\top$ is a projection only then. Every positive variance mode is attenuated smoothly by $\alpha/(\lambda_i + \alpha) \in (0, 1)$, hardest on the widest nuisance directions, while the orthogonal nullspace ($\lambda_i = 0$) stays at weight 1: components are **shrunk, not deleted**, so signal lying along a noisy axis is retained at reduced weight. We set $\alpha = \lambda_{16}$ (`M1_ridge_lam16`) throughout, spanning the $S - 1 = 13$ subcategory degrees of freedom plus the leading style axes with zero learned parameters, gradients or search.

**The threshold.** $L$ is a **declared leak budget**, not a measured property. With $m = \lfloor L \cdot \text{rows}(X_n) \rfloor$, two rules:

$$\text{R2:} \quad t = \min\big\{\, s \in \text{TRAIN} : \#\{X_n \ge s\} \le m \,\big\} \quad (6)$$

$$\text{R3:} \quad t = r_{m+1} + \varepsilon \quad (7)$$

where $r$ are the sorted REF scores. R2 can only place $t$ on one of the TRAIN scores, and we show a case where the operating point we wanted fell in a **gap in that staircase** — 0.0323 of score with no candidate in it. R3 has a finer grid and cannot become vacuous, because the constraint itself sets $t$. $\varepsilon$ **must exceed the model's measured live-versus-offline reproducibility bound**: a one-ULP version failed, its binding certificate surviving neither live scoring nor a change of BLAS reduction order (§8).

A stated rule can also fail to be the thing that produced the threshold. If $\min(\text{TRAIN}) > r_{m+1}$ then every candidate was feasible, $t$ fell onto the lowest TRAIN score, and $\#\{X_n \ge t\} = m$ is a coincidence: we call such a cell VACUOUS, report the distinction for every cell, and do not ship one.

**At serve time.** $\text{score}(x) = \langle x/\|x\|, c\rangle$, fire iff score $\geq t$. The decision is taken once at prefill, stored as a per-sequence boolean, and not revisited per token.

### 2.4 Choosing the read layer

Two constraints, one hard and one empirical.

**Hard: the read must be strictly below the edit band.** $\max(\text{read}) < \min(\text{edit})$ is asserted at construction and an in-band read is refused. This is the constraint the title calls *before* the edit: a lower layer index means the read happens earlier in the forward pass. Everything in §6 rests on it — every layer the gate reads is unmodified, so $h$ at the read site is bit-identical in both arms, the decision cannot depend on the edit, and the edit can be applied to the same pass that produced the decision. The selected read layers are **L8, L11 and L10**, 2, 2 and 1 layers below each model's band, and each model uses that same single read layer for both the cyber and bio gates. That harm detection sits upstream of refusal execution is published as anatomy: Chu et al. [6] characterise a safety circuit in which *"Harmful Detection Heads"* precede the *"Refusal Heads"* that turn the signal into a refusal.

**Empirical: select on out-of-corpus data, by the operating statistic.** Within the admissible layers we select on a held-out half of an independently assembled corpus, ranking layers by **leak at matched coverage**. Two rules we explicitly reject. *AUC*: it is not the statistic the gate operates on, and it rises nearly monotonically with depth, so an AUC argmax always returns the deepest admissible layer. *Any in-corpus argmax*: selecting Phi's read layer by leak at matched coverage over 100 paired in-corpus resplits chose L5 over L11–L12 by $+3.45$ leaks per 378, bootstrap $[+2.72, +4.16]$, Wilcoxon $p = 4.6 \times 10^{-12}$ — and on a fresh corpus the sign reversed at every point tested, the out-of-corpus interval at matched coverage being $[-21, -1]$, which does not overlap. More seeds could not have helped: every resplit carries the same ratio of dimensions to rows, so resplit stability is structurally blind to this class of bias. The shipped layer on Phi is L11 (with L12 also verified live on the bio gate). The full account is in the long version of this paper and in `records/phi-oocsweep-2026-10-06/`.

**Where the depth trend sits in the probing literature.** Linear separability rising with depth is the canonical probing finding [2], though the richest representations are reported at mid-depth [27]; prompt domain is decodable from prefill hidden states and has been used to route between models [12]; and for adversarial intent specifically, *"surface form appears early, while pragmatic intent consolidates later"* [21].

## 3 Related work

**Prior building blocks.** Our design connects several lines of prior work:

- **Domain-specific refusal removal.** Hadetskyi et al. [8] remove refusal on Kimi K2 for the cybersecurity domain via multi-objective search without a runtime gate; Palette [29] selectively relaxes refusal on authorized target domains while preserving safety elsewhere.
- **Terminology.** The phrase "conditional directional ablit-eration" is used in community code (`apostate/diode`, [3]); we say **gated** to denote a single pre-band routing decision over a rank-1 weight edit.
- **Runtime gating.** CAST [17] reads a condition at layer 7 for activation steering; Residual Paving [11] routes from layers 1–4 and intervenes at later layers; CLEAR [30] drives a safety adapter from a hidden-state gate; SafeMoE [10] routes with a learned gating network; and Tripwire [36] realises its clamp as a detector-gated intervention.
- **Contrast-set composition.** Petrov et al. [22] show that the composition of the contrast baseline determines whether a refusal direction separates cleanly.

Six properties, counted generously (a partial counts as a half): (1) closed-form direction; (2) gate **below** the edit; (3) rank-1 weight edit; (4) narrow band; (5) compliance; (6) category routing. Nothing published reaches five (Table 1).

**The gap is properties 2 + 3 + 5 together.** Each pair exists. Hadetskyi et al. [8] score highest and have no gate at all. Residual Paving [11] is the closest on architecture and deserves the credit for it: it forms a boundary feature from residual states in $L_R = \{1, \ldots, 4\}$ and intervenes at later layers $L_I$, which is the read-early/edit-late separation this paper is built on. It differs in what is gated and how the gate is obtained. What is gated is an added set of prompt-conditioned bottleneck residual experts aimed at a *"safe reframe"* that its authors state is **"not a direct-compliance target"**, with the backbone left unchanged — rank-1 orthogonalisation is absent from the paper. How the gate is obtained is gradient descent: the router is fitted *"to the edit-versus-keep boundary on cached early residual states with weighted BCE"*. Their own diagnostic then makes that router the headline: substituting a held-out oracle label for the learned scalar gate raises their keep-side score on *every* one of six backbones, by a median $+12.9$ pp and as much as $+50.7$, and learned benign preservation ranges from $57.2\%$ to $97.9\%$ across backbones while oracle routing reaches $\geq 99.9\%$ on all of them. They conclude that *"learned route selectivity remains the limiting factor"*. That conclusion is the motivation for the present paper: we keep their layer separation and replace the trained router with a closed-form one. We report no head-to-head — the backbones, the edit sets and the metrics all differ —

Table 1: The six properties against the nearest published and community work. Y = yes, N = no, P = partial. Qualifications that would not fit a cell are given in the notes below, verbatim from the draft.

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Hadetskyi et al. [8] | Y | **N**[a] | Y | P[b] | Y | P |
| CAST [17] | P | P[c] | **N** | N | P | Y |
| Residual Paving [11] | N[d] | **Y**[e] | N[f] | P | **N**[g] | N |
| `apostate/diode` [3] | Y | **N**[h] | N[i] | P | Y | N |
| RepIt [26] | P | **N**[a] | N[j] | N | Y | Y |
| Palette [29] | N[k] | **N**[l] | P | N | Y | Y |

[a] no gate at all. [b] spread over 25–95% of depth. [c] condition at L7; no edit, so no "below". [d] learned. [e] router at layers 1–4, ahead of a later-layer edit. [f] added bottleneck experts. [g] a *safe reframe*, which its authors state is not a direct-compliance target. [h] gate co-located inside the edited layers. [i] gated MLP neuron. [j] activation steering, not a weight edit. [k] multi-objective search. [l] no runtime gate.

so this is a difference in construction, not a demonstrated improvement. CAST [17] has gate and category but **"no weight update"**, so nothing exists for the gate to sit below. Two further near neighbours: LoMC [13] matches our operator — a rank-one layer-wise correction applied only inside a compact edit support — but its *"support-gated"* gate is an offline mask over *weights*, *"updates outside $\mathcal{S}$ are set to zero"*, not a read of the prompt, and its direction is a single harmful-versus-benign one rather than a category; and Alagharu et al. [1] do pair per-category refusal directions with *"a lightweight probe that determines whether to steer"*, but the categories exist only because the model was fine-tuned to emit categorical refusal tokens, the intervention is a steering vector, and the direction of travel is *increasing* refusal.

**There is also a published argument against the design, and it is explicit.** C-$\Delta\Theta$ [16] confines a weight edit in order to *eliminate* the gate rather than to enable one: *"conditional variants improve selectivity by gating when steering is applied but still retain an inference-time control path"*, and what it ships instead is *"no inference-time hooks, no per-request gating"*. Three further results say the gate is the fragile part: Residual Paving [11] (*"the oracle route is non-deployable"*), Bosco and Srinivasan [5] (a detector-gated intervention *"only matches a trivial rule"*), and Xun [33] (a calibrated gate OOD-inverts and *"silently reduces the gated pipeline to its base model"*). We take that as the standard this design has to meet, which is why §6 is a verification section rather than a table of rates.

# 4 Experimental setup

**Models.** Three architectures, in Table 2. Gemma is the only MoE, so 7 of its 21 tensors are fused `experts.down_proj` stacks; the implementation proves its reconstruction of each stack's input against that stack's own forward before hooking it and declines any stack exceeding tolerance, because a mis-indexed fused edit is a wrong edit rather than a smaller one — Shi et al. [25] report that module-name matching drops such stacks and *"nothing raises an error"*. No weight is modified on disk in any experiment; the edit is applied as forward hooks on unmodified weights, verified byte-identical before and after by shard hash, inode and link count.

**Corpora.** No prompt text is reproduced anywhere in this paper (§9). The FIT set is `pool681`: 1426 rows assembled from advbench / eval177 / `harmful_behaviors` test — 692 harmful over 15 categories, 554 harmless, 180 excluded paraphrase rows. The OOC set is 3216 fresh rows: CySecBench 1350 (cyber), CatQA 469 (10 general harm categories), plus ICS, imperative and replication arms and 40 convention-control rows. A third set, HarmBench [20] with per-category labels, exists for Gemma only (197 rows). Disjointness between FIT and OOC: **0 exact matches** at three normalisations, token-set Jaccard nearest neighbour **max 0.438** with nothing at $\geq 0.5$. The bound is stated: exact matching cannot catch a low-overlap paraphrase. Every fresh capture carries a convention control — a subset of FIT rows re-captured and compared against the stored activations, at cross-run distances of 4.12e-05, 1.38e-05 and 1.77e-05 against within-run bf16 floors of 7.44e-05, 1.21e-05 and 1.93e-05, with a wrong-layer sentinel $1350\times$ and $4517\times$ those distances on the two dense models. Without this, a convention drift would present as a model difference.

**Judging.** A three-way judge — REFUSED / DEFLECTED / COMPLIED — on an independently served model, rubric fixed across every arm. Three-way-or-finer response taxonomies are established [24, 35]. A keyword refusal count cannot distinguish *"I won't help"* from *"I won't help, but here is how you would do it"*, and those are different outcomes. That failure has been measured: keyword refusal precision of 11.2% across six abliterated models [34]; a Spearman correlation of $-0.394$ between string matching and human rankings [28]; substring ASR moved by up to 30 points by the generation budget alone [20]. Such marker lists are nonetheless the field's default [32]. The judge is probed before use on a clear refusal, a clear compliance and a clear deflection; judge noise, measured on byte-identical text across two runs, is **1 label in 138**.

**A reply either refuses or it answers.** The judge reports three labels, but the headline counts two states: *answered* means the reply is not a refusal. The prior work's keyword

Table 2: The three models, the edit each receives, and the result of applying that edit *unconditionally* — same 100 prompts, same judge, same decoding, no gate. The read layer is strictly below the band in every row, which is what §2.4 requires. OFFLINE.

| model | arch | params | layers | hidden | edit band | read | tensors | $K$ | original refused | edited refused | KL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| `gemma-4-26B-A4B-it` | MoE | 26B | 30 | 2816 | L10–16 | L8 | 21 | 2.0 | 97/100 | **1/100** | **0.0849** |
| `Phi-4-mini-instruct` | dense | ∼4B | 32 | 3072 | **L13–15** | L11 | **6** | 2.0 | 99/100 | **0/100** | **0.0745** |
| `Llama-3.1-8B-Instruct` | dense | 8B | 32 | 4096 | L11–22 | L10 | 24 | 1.5 | 94/100 | **1/100** | 0.0905 |

Full identifiers: `google/gemma-4-26B-A4B-it`, `microsoft/Phi-4-mini-instruct`, `meta-llama/Llama-3.1-8B-Instruct`. The KL column mixes two harnesses and always did; the ordering it appears to give is not one we claim (§5).

detector cannot make the third distinction and scores an evasive reply as a successful removal, so the two-state reading is the comparable one. We apply it identically to the cyber and non-target harmful sides — it helps us on one and hurts us on the other — and we publish the three-way counts so anyone can recompute.

**We count only the prompts the original model refused.** The original model already answers 16/64 Gemma, 16/64 Phi and 13/64 Llama CySecBench prompts on its own, because CySecBench phrases requests analytically. On those prompts the edit has nothing to remove, so they cannot show its effect. Earlier drafts subtracted a baseline; dropping the prompts is cleaner than correcting for them.

# 5 Results: the edit

With no gate, the edit removes refusal almost completely on all three architectures at KL between 0.0745 and 0.0905 (Table 2). **Phi removes refusal completely with 6 tensors at 9.4% of depth.** For contrast, Hadetskyi et al. [8] spread their edit across 25–95% of depth and report $100\% \rightarrow 7\%$.

**One ordering we do not claim.** The KL column mixes two harnesses: Gemma's cell comes from one driver at batch size 20, Phi's and Llama's from another at batch size 8 which also differs in `add_special_tokens`. We bounded that term by re-measuring Gemma's edit at batch 8: **0.0945,** $+11.3\%$ — larger than the Gemma–Llama gap, and it flips their order ($0.94\times$ Llama at batch 20, $1.04\times$ at batch 8). So **Gemma versus Llama is not rankable from these numbers** and we do not rank them. Keyword refusal counts are identical (33/100) at both batch sizes, so only the first-token KL moves.

The refusal count this literature reports — and that we adopt for comparability — treats a deflection as a successful removal, so it flatters every method including ours. The full split as REFUSED/DEFLECTED/COMPLIED is Gemma **1/30/69**, Phi 0/5/95, Llama 1/3/96: thirty of Gemma's 99 non-refusals withhold substance. A reader who wants the strict reading can take the third number, and §6.3 shows our gate conclusion holds either way.

Controls, all three models: the expected tensor count was hooked exactly with **0 direction-keyed tensors unreached**; the always-on version computed **no** gate score, so the classifier is absent from this measurement; the original version attached 0 hooks and fired 0; and **text differs on 100/100, 100/100 and 99/100** prompts. Gemma's original version matched an independent harness **byte-for-byte on 100/100 prompts**, which demonstrates it is the unedited model's own output. Record: `records/abl-control-2026-10-07/`.

# 6 Results: the gate confines the edit

Three versions of each model answer the same prompts: **original**, no edit at all; **always-on**, the edit applied to every prompt with no gate; and **gated**, the classifier deciding prompt by prompt. One measurement runs on every prompt — does the gate fire? — because that is scoring only, so it covers the whole fresh pool: 675 cyber prompts and 234 non-target harmful ones.

## 6.1 Firing

**Three architectures land in one tight band: 78–82% of target prompts routed to the edit, for 7–9% leak** (Table 3, cyber-gate row; on the 637-row WMDP_CYBER test set, the same three gates fire on 602 / 506 / 600 = **94.5 / 79.4 / 94.2%**). Three different bands, three independently fitted directions, one prompt set (`prompts_sha256=11faeab0...`). We do not pool the three rows: they sit at three different thresholds, and an average over operating points is not a quantity any served model exhibits.

**The firing rate is also the answer rate.** We report no separate column for whether the model then answered, because on the prompts where it could matter the two are the same number: the gate decision and the judged outcome agree on **294 of 294** (§6.2). On the 64-prompt fresh cyber generation sample, gated compliance reaches **55/64 (85.9%) on all three models** (up from 16/64, 16/64 and 13/64 unedited, delivering 84.8%, 84.8% and 85.7% of the forced-edit ceiling), while preserving 10/10 compliance on the benign generation control. The edit applied with no gate answers **100%** of the refused prompts, so the gate is withholding a capability that is fully present, not dividing up a smaller one.

**All three rows are live**, served by our own HTTP server with one forward pass per prompt (`records/cyber-m1-live-serve-2026-10-15`). Across all 2875–2876 scored rows per model, live and offline decisions agree on **99.86%, 99.93% and 99.90%** (4, 2 and 3 decision flips total), and re-submitting the final 24 rows (P10 replay control) yields **24/24 bit-identical IEEE-754 hex scores** ($\max|\Delta| = 0$) on every run. Measured $\max|\text{live} - \text{offline}|$ across the full sweep is 6.92e-03 on Gemma, 1.01e-02 on Phi and 1.02e-03

Table 3: **The complete picture.** Two gates (`M1_ridge_lam16`, $\alpha = \lambda_{16}$), three prompt sets, three architectures; cells read Gemma (L8) / Phi (L11) / Llama (L10). The diagonal (bold) is coverage, the rest is leak. Counts first, rate second. Every cell LIVE.

| gate \ prompts | cyber, 675 CySecBench | bio, 637 WMDP_BIO test | non-target harmful, CatQA |
|---|---|---|---|
| **cyber gate** | 526 / 549 / 551 = **77.9 / 81.3 / 81.6%** | 5 / 2 / 33 = 0.78 / 0.31 / 5.18% | 16 / 22 / 17 = 6.8 / 9.4 / 7.3% |
| **bio gate** | 3 / 0 / 2 = 0.44 / 0.00 / 0.30% | 633 / 630 / 632 = **99.4 / 98.9 / 99.2%** | 15 / 8 / 10 = 6.4 / 3.4 / 4.3% |

The non-target harmful column is the CatQA TEST-half aggregate *excluding* each gate's own target column, $n = 234$ (Phi 235) for the cyber gate and $\sim$234 (Phi 235) for the bio gate. Each model uses one shared read layer across both gates: L8 on Gemma, L11 on Phi, and L10 on Llama (serving Phi's bio gate at L12 instead gives 629/637 = 98.7% bio coverage, 0/675 = 0.00% cyber leak and 11/235 = 4.7% CatQA leak). Shipped '.pt' thresholds $t$ (`op["thresh"]`): cyber gate 0.1134911512 (Gemma L8, `iso_leakage`), 0.0530322825 (Phi L11, `iso_leakage`), 0.0689013176 (Llama L10, `in_fit_R3`); bio gate 0.0580110430 (Gemma L8, `R3U140`), 0.0422080616 (Phi L11, `R3U140`; 0.0419398590 at L12), 0.0852456303 (Llama L10, `R3U140`). Offline Phase-0 cyber delivered counts stored in `meta["delivered"]` are 528 / 549 / 550 (CatQA leak 16 / 21 / 17). Clopper–Pearson 95% intervals — cyber gate on cyber: [74.6, 81.0], [78.2, 84.2], [78.5, 84.5]; on non-target harmful: [4.0, 10.9], [6.0, 13.8], [4.3, 11.4]; cyber gate on bio: [0.26, 1.82], [0.04, 1.13], [3.59, 7.20]; bio gate on cyber: [0.09, 1.29], [0.00, 0.55], [0.04, 1.07]; bio gate on bio: [98.4, 99.8], [97.8, 99.6], [98.2, 99.7]; bio gate on non-target harmful: [3.6, 10.4], [1.5, 6.6], [2.1, 7.7]. Across all 2875–2876 scored rows per model on the cyber gate, live and offline decisions agree on 99.86–99.93% (4, 2 and 3 flips total; on CatQA, Phi's 22 live vs. 21 offline is a single row 3.3e-05 below $t$, while Gemma's 16 and Llama's 17 have 0 flips). On the bio gate, Phi L11 has **0 flips across all 3266 rows**, while Gemma L8 and Llama L10 have 3 and 2 flips across 3265 rows. Records: `records/cyber-m1-live-serve-2026-10-15` and `records/bio-m1-live-serve-2026-10-15`.

on Llama (with mean absolute drift 4.21e-04, 1.51e-04 and 1.68e-04, and $|\text{AUC}_{\text{live}} - \text{AUC}_{\text{offline}}| \leq 2.8\text{e}{-04}$ across all 24 model–pair evaluations). At Gemma's in-fit R3 threshold ($t = 0.102134$), live coverage rises further to **558/675 (82.7%)** at 18/234 (7.7%) CatQA leak; at its in-fit R2 threshold ($t = 0.122286$), CatQA leak drops to **12/234 (5.1%)** and bio leak to **2/637 (0.31%)**.

**A note on which prompts count.** The original model already answers some cyber prompts by itself. Those prompts stay in Table 3: the gate's job is to recognise the topic, not to predict what the base model would have done, and the gate fires on them at a comparable rate — pooled across the 64-prompt cyber subset, 90.7% (68/75) against 82.9% (97/117) on the prompts the original refused. They are excluded only from §6.2, which asks whether the *edit* changed the reply; there a prompt with no refusal to remove cannot show an effect.

## 6.2 On refused prompts, the gate bit is the whole story

Here we ask whether the edit changed the reply, so the prompts are restricted to those the original model refused on the 138-prompt served subset.

Table 4: Gate bit against judged outcome, on prompts the original model refused (`M1_ridge_lam16`). The two middle columns are empty. LIVE.

| | | fired | | held | |
|---|---|---|---|---|---|
| | $n$ | answ. | refused | answ. | refused |
| Gemma cyber | 22 | 15 | 0 | 0 | 7 |
| non-target harmful | 58 | 5 | 0 | 0 | 53 |
| Phi cyber | 45 | 39 | 0 | 0 | 6 |
| non-target harmful | 58 | 7 | 0 | 0 | 51 |
| Llama cyber | 50 | 43 | 0 | 0 | 7 |
| non-target harmful | 61 | 6 | 0 | 0 | 55 |

**294 prompts, 294 agreements, three architectures.** The two middle columns are empty (115 fired and answered; 179 held and refused). Nothing was answered without the gate firing, and nothing that fired was refused; whether the model answers is decided by one bit.

## 6.3 The strongest evidence is byte identity

On every prompt where the gate does **not** fire, the served output is the unedited model's output exactly.

Table 5: Byte identity on withheld prompts — the two *harmful* test sets, cyber and non-target harmful. No harmless prompt appears in this table. LIVE.

| | gated = original | always-on ≠ original |
|---|---|---|
| Gemma | 9/9 and 58/58 | on all of them |
| Phi | 10/10 and 57/57 | on all of them |
| Llama | 8/8 and 58/58 | on all of them |

The edit would have changed every withheld prompt, and the threshold is what stopped it. This is not a rate; it is identity, and it is the property §2.4's constraint was imposed to obtain. In addition, live 3-arm `/compare` and `/generate` verification across all discordant and control rows on the 138-prompt subset (113 live generations across the three models in `records/cyber-m1-live-serve-2026-10-15`) confirmed 100% `sha256` byte identity between the served gated output and the forced-edit branch whenever the gate fired, and 100% byte identity with the unedited model whenever held.

## 6.4 Two bookkeeping notes

**Why R2 can stall on a staircase while R3 does not.** An in-fit leak budget is an integer $m$ over 140 reference prompts, and R2 restricts $t$ to the discrete set of 100 TRAIN scores. When few TRAIN prompts score in the tail of the reference distribution, R2's candidate grid becomes coarse: on Phi ($m = 14$), R2 jumps across a score gap from $L = 0.10$ to carry a realised CatQA leak of 15/235 (6.4%) at 415/675 cyber

coverage, whereas calibrating $t$ at the matched leak budget (`iso_leakage`, $t = 0.053032$) routes **549/675 (81.3%)** at 21/235 offline (22/235 live) CatQA leak and **2/637 (0.31%)** bio leak. On Llama ($m = 17$), the in-fit R3 rule ($t = 0.068901$) and `iso_leakage` ($t = 0.068867$) coincide to within $3.4 \times 10^{-5}$ and produce the exact same live counts (**551/675** cyber and **17/234** CatQA).

**Why deterministic routing makes threshold sweeps exact.** Because the gate reads only layers the edit never touches, the gated output under greedy decoding is a deterministic function of the single prefill gate bit: it equals the forced-edit completion when score$(x) \geq t$ and the unedited completion when score$(x) < t$, verified byte-for-byte via `sha256` across all 113 live `/compare` and `/generate` evaluations in `records/cyber-m1-live-serve-2026-10-15`. Moving $t$ or updating $c$ therefore switches each prompt cleanly between two completions the judge has already scored, with discrepancy possible only on prompts within live-versus-offline float jitter of $t$.

# 7 Results: a second target category

The cyber gate is not the only gate the recipe produces. Table 3 is the complete picture: two gates, three prompt sets, three architectures, every cell served live. The diagonal is coverage, where high is the goal; the off-diagonal is leak, where low is the goal. Each gate fires on most of its own category and on a few percent of the other's, on every architecture. The off-diagonal is not a shared block of ambiguous prompts: 34 of the 637 bio prompts fire some cyber gate, the three models' sets are largely disjoint, and only two prompts fire all three.

**The cyber-on-bio cell is the only selection-clean number in the table.** No WMDP row entered the cyber gate's fit, its layer choice or its threshold. The other cells were each chosen at an operating point selected with reported surfaces in view, so this one carries the claim most cleanly — and the cyber gate in fact leaks *less* onto bio than onto the non-target harmful pool, at $0.11\times$ / $0.03\times$ / $0.71\times$ of its 6.8 / 9.4 / 7.3% there (and under in-fit R2, cyber-on-bio leak drops to **2 / 1 / 6 = 0.31 / 0.16 / 0.94%**).

**Bio beats the cyber gate on both numbers on all three models — 98.9–99.4%** coverage (633 / 630 / 632 of 637; 629 = 98.7% at Phi L12) on out-of-corpus target prompts at **0.00–0.44%** leak (3 / 0 / 2 of 675) on our cyber set and **3.4–6.4%** (15 / 8 / 10) on our other-harm set, against the cyber gate's 78–82% at 6.8–9.4%. It fires on **0 / 0 / 0 of 637 (0.00%)** same-collection *cyber* prompts (`wmdp_cyber_test`), so it is reading the category and not the corpus. Moreover, on the external 498-prompt SOSBench bio benchmark, the same three gates fire on **480 / 473 / 443 = 96.4 / 95.0 / 89.0%** (477 = 95.8% at Phi L12), and on its 310-prompt keyword-free bio subset (containing zero WMDP_BIO training token stems) they fire on **297 / 288 / 267 = 95.8 / 92.9 / 86.1%** (291 = 93.9% at Phi L12), clearing the pre-registered 234/310 (75.5%) concept-generalisation bar on every architecture while firing on only 22 / 7 / 3 of 612 PKU cybercrime prompts (0.5–3.6%).

**Two threshold defects had to be fixed first, and the fix was arithmetic rather than search.** Both earlier bio attempts failed at 90–98% leak while carrying a sound direction. *(i) Corpus:* a budget spent on reference rows from one corpus does not buy the same rate on another — on Llama, 14/140 by construction against 371/675 realised. The reference rows must come from the corpora the leak is measured on. *(ii) Budget:* $m = 14$ on 140 reference rows has a Clopper–Pearson upper bound of **16.21%**, so calibrating against a 10% target was always consistent with a realised 16.2%, and every cell that missed did so inside that band (10.2–15.8%). $m = 6$ is the largest budget whose upper bound stays at or below 10%. That is derived from the interval arithmetic alone, with no score involved.

**Confirmed live, and 100% on the clean validation surface.** Because the bio operating point was selected with test surfaces in view and WMDP_BIO's train and test splits share duplicate stems, coverage was also read live on the **127 validation rows (`wmdp_bio_val`) that have zero overlap with train and entered no fit, no threshold and no grid cell**: live coverage is **127 / 127 / 127 = 100.0 / 100.0 / 100.0%** (Clopper–Pearson 95% interval [97.1, 100.0], and 118/118 = 100.0% after exact deduplication) across all three models (and 127/127 at Phi L12 as well). That closes the selection question for coverage; every leak surface did enter the grid.

**Live numerical fidelity and threshold binding.** Across all 2971–2972 offline-referenced rows in the bio live sweep, max $|\text{live} - \text{offline}|$ is **6.39e-03 on Gemma L8, 5.48e-03 on Phi L11 (3.76e-03 on Phi L12) and 9.68e-04 on Llama L10** — all strictly inside the pre-declared 7.17e-03 jitter band ($0.89\times$, $0.76\times$ and $0.14\times$). Phi L11 exhibits **0 decision flips across all 3266 rows** and reproduces the exact in-fit reference count $\#\{\text{REF} \geq t\} = 6/140$ live. Because R3U140 places $t$ one ULP (1.39e$-$17) above the $(m{+}1)$-th reference score $r_7$, a single reference boundary row on Gemma L8 and Llama L10 drifts by 1.7e$-$04 and 2.4e$-$05 across $t$ under live `bfloat16` scoring (yielding 7/140 on REF at R3U140, and 6/140 on Gemma under R3$\varepsilon$U140 with $\varepsilon = 10^{-3}$). No reported test cell depends on the fit reference set realising $m$ to the integer, so we treat the in-fit budget as nominal and rely on the out-of-corpus coverage and leak margins instead. Record: `records/bio-m1-live-serve-2026-10-15/`.

**Two candidates the recipe does not deliver, and one it cannot score.** Influence ops and fraud fire on 60–88% and 73–81% of their own targets but leak on 15.3–22.5% and 19.9–28.4% of non-target harmful prompts — five to ten times the cyber gate — and are disqualified on that column (`records/fraud-gate-2026-10-07/`, pre-registered before any direction was fitted). A fourth candidate, hate, is not reported here at all: its own-target coverage could not be established, two hate collections bracketing it between 18% and 100% for the same gate at the same threshold, and the full account is in the long version of this paper and in `records/hate-gate-2026-10-07/`.

# 8 Limitations

**Two corpora.** Every interval above bounds row sampling within one corpus. Nothing here bounds corpus-to-corpus variation, and two corpora cannot. Only Gemma has a third.

**A declared leak budget permits more than its label says.** The budget is an integer count $m$ over a finite reference set, so what it guarantees is an interval, not a rate. $m = 14$ on 140 reference rows has a Clopper–Pearson 95% upper bound of **16.21%**. The cyber budgets of $m = 11/14/17$ permit **13.6 / 16.2 / 18.7%** respectively. The cyber gate's observed 6.8–9.4% leak is therefore an outcome, not a guarantee of its budget, and a reader should not treat the declared $L$ as a bound on the realised rate. The bio gate's $m = 6$ has an upper bound of 9.09% — the tightest budget in this paper, and the only one whose label survives the interval arithmetic.

**Live decisions can differ from offline ones.** On the 2875–2876 scored rows of the cyber gate sweep (`M1_ridge_lam16`), measured max $|\Delta|$ between served and stored scores is 6.92e-03 on Gemma, 1.01e-02 on Phi and 1.02e-03 on Llama (mean 4.21e-04, 1.51e-04 and 1.68e-04), producing 4, 2 and 3 decision flips across the entire 2875-row surface (99.86–99.93% agreement); on the 3265–3266 rows of the bio gate sweep, max $|\Delta|$ is 6.39e-03, 5.48e-03 and 9.68e-04, producing 3, 0 and 2 flips. All per-model jitter figures are **sample maxima rather than bounds** (§6.1). Relatedly, one-ULP threshold certificates are fragile at float precision: placing $t$ a single ULP above $r_{m+1}$ cannot immunise against $10^{-4}$ live `bfloat16` reduction drift unless a finite margin $\varepsilon$ is added (§7). All served thresholds in `records/cyber-m1-live-serve-2026-10-15` and `records/bio-m1-live-serve-2026-10-15` are stored and verified at full IEEE-754 float64 precision.

**Capability retention.** This is measured on one model. Gemma has a blind, position-randomised pairwise arm over 50 prompts (McNemar $p = 1.00$ against the original). Phi and Llama have 3 and 8 benign prompts, and benign compliance *fell* on both (Llama $9 \to 6$, Phi $8 \to 7$, into DEFLECTED not REFUSED). KL cannot substitute: this project measured a build at half the KL judged *worse*.

**Gemma self-judges.** The judge is the same model family as the Gemma checkpoint under test. The Phi and Llama rows do not carry this.

**A withdrawn claim.** Our contrast with Hadetskyi et al. [8] rested on a result recorded as *"not reproduced. Zero selectivity."* An audit finds that arm differs from their method on six points, four of them load-bearing: a dense 8B rather than a 1T MoE, where scale is part of their stated mechanism; a 1-harmful-plus-4-harmless contrast kit rather than their prompt sets per domain; CBRN as the judged target rather than their cyber; and CBRN against fraud as the selectivity pair rather than cyber against explicit content, with no explicit-content prompt in the probe. Edit span and operator normalisation also differ. Independently, every arm in that sweep sat at a corner of the dose dial (0/0/0 or 88/92/82) while their claimed state is partial, so the design could not have detected selectivity had it been present. **All we assert is what was measured: a single-prompt domain-matched kit on Llama-3.1-8B is fully functional and completely non-selective —** $94/100/92\% \to 0/0/0\%$ **on 96 prompts. We make no claim either way about whether their result holds.** Record: `records/replication-audit-2026-10-07/`.

**We do not claim that corpus-driven selectivity fails.** We reproduced RepIt [26] on Phi-4-mini with their released code and zero source modifications: target ASR **0.8023** against their published **0.8136** (McNemar $p = 0.875$), held-out selectivity 3.55 baseline-adjusted. Their raw difference-of-means direction already discriminates at 1.98 there, because the corpus was purpose-written and single-concept — the same lever Hadetskyi et al. [8] claim. Our contribution is the gate and its position, not a negative result about contrast-set design.

**Llama is a measurement arm only.** No Llama artifact is distributed, under the licence constraint for modified Llama weights.

**Prepared but unrun.** A CBRN gate on 670 WMDP positives is fully prepared and was not executed — splits audited disjoint, dry run passed, positive controls reproduce the shipped constants, and the GPU stage never started. We report no number for it.

**What this paper leaves out.** Two further results are omitted here for space and kept in the long version: an evaluation-protocol ladder showing that the same classifier scores 96–98% target coverage when test rows are drawn from the corpus it was fitted on and 38–66% when they come from an independently assembled corpus, and the hate category discussed in §7. The first is a contribution about evaluation methodology rather than about the gate; every coverage figure reported here is already on the separated side of it.

# 9 Reproducibility and data handling

The verified proof-of-concept implementation, runtime gate server and shipped operating-point kits are available at `https://github.com/kncshw/resgate`. Every number traces to a record directory under `ablation/records/`, each containing the as-run scripts, the logs, the per-row outputs and a summary JSON with sha256 for every input; the record names are given beside the results they back, and the long version of this paper carries the complete index. Operating points ship as compact `.pt` files — a direction blob and an operating point carrying its own provenance. Positive controls are mandatory and gated: no result is reported unless the implementation first reproduces six published constants to float64.

**Data handling.** The WMDP prompts used for the bio work were shared privately by the RepIt authors, whose paper records them as not publicly available pending gated access. They are not redistributed, no weight, direction blob or operating-point file from the bio arm ships, and **no prompt**

**text from that collection appears anywhere in this paper**. Prompt sets are described by name, source and count only. Every artifact bound for the bio record was screened against 56,004 distinct four-word needles drawn from all 2954 author prompt strings; escalating to 58,362 eight-word needles over all 121 files in `records/bio-m1-live-serve-2026-10-15/` gives 0 hits.

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